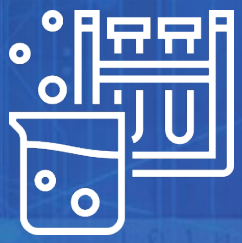
Lab Assignment & Solution



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Cybersecurity Professional Program

Ethical Hacking

Web Application Security Fundamentals

EH-09-LS1

Personal HTTP Server

Note: Solutions for the instructor are shown inside the green box.

Lab Objective

Become familiar with HTTP headers, Apache service control, configuration files, and a web development inspector tool.

Lab Mission

Customize a personal HTTP server with a connection port and HTTP header.

Lab Duration

15–45 minutes

Requirements

- Knowledge of Kali OS.
- Basic knowledge of networking.
- Basic knowledge of HTTP.

Resources

- Environment & Tools
    - VirtualBox
        - Kali Linux
            - Apache2
            - Browser

Textbook References

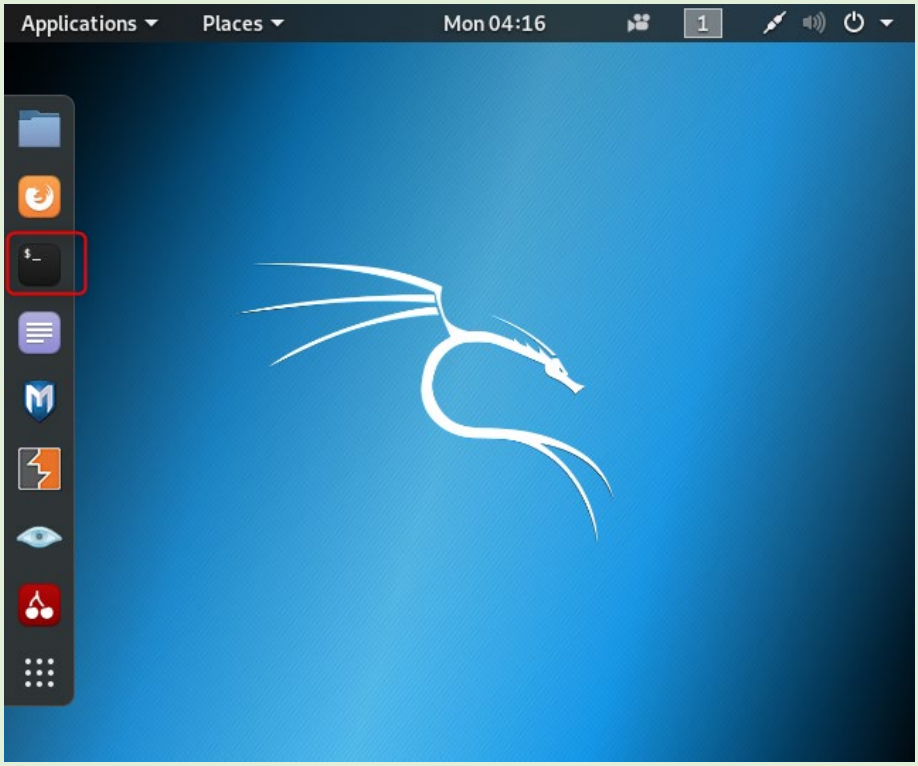
- Chapter 9: Web Application Security Fundamentals
    - Section 1: Understanding Web Technologies

Lab Task 1: Basic Apache Configuration

In this task, you will perform a basic custom configuration for your Apache web server and change its default listening port.

1. Start your Kali machine.
2. Start the Apache service and verify that it is active.
3. Change the Apache port configuration file to listen on port 9191. Now restart the Apache service.

Start a terminal session by clicking the terminal icon in the menu bar.

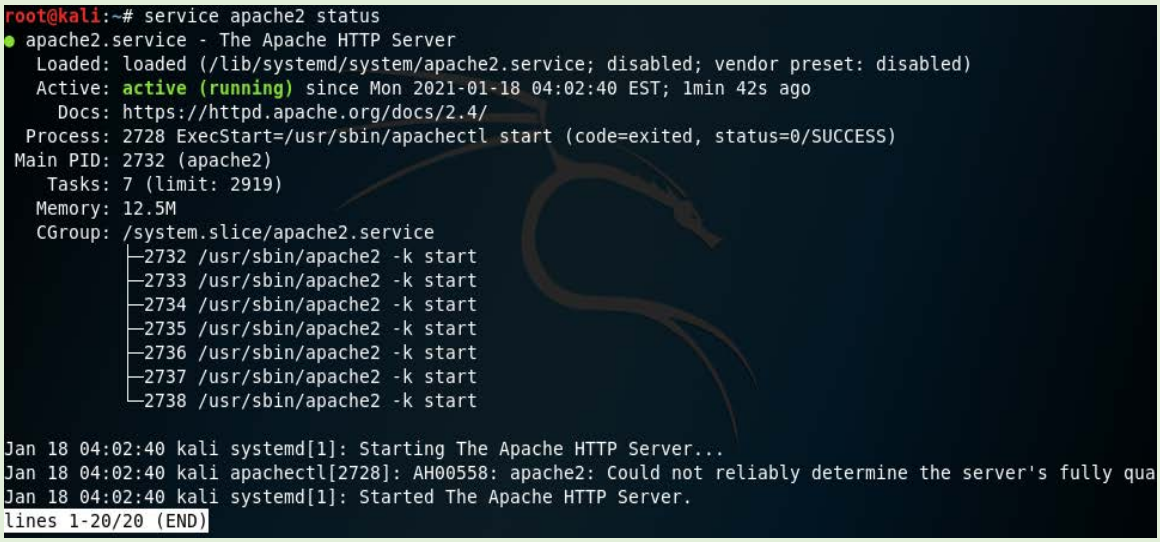


Type **service apache2 start** in the Kali terminal, and press Enter.

```
root@kali:~# service apache2 start
root@kali:~#
```

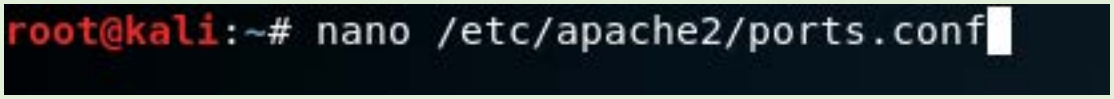
Type **service apache2 status** to check the status of the Apache2 service.

Press **Q** to exit the status process.



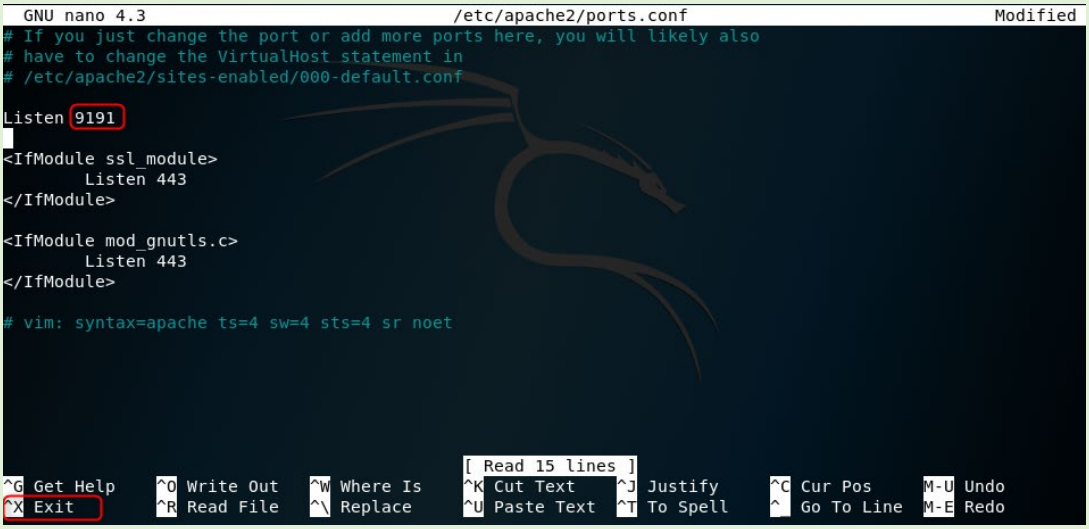
Open the Apache port configuration file and change it to listen on port 9191.

Type **nano /etc/apache2/ports.conf** to edit the configuration file.



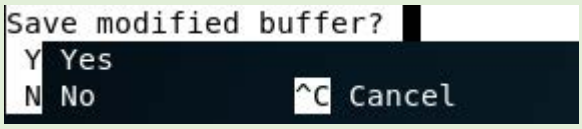


Use the down arrow key to scroll to “Listen 80”, and then use the right arrow key to move to “80”, and change it to “9191”.



Press CTRL+X to exit the Nano editor.

Press Y (Yes) to save the changes.

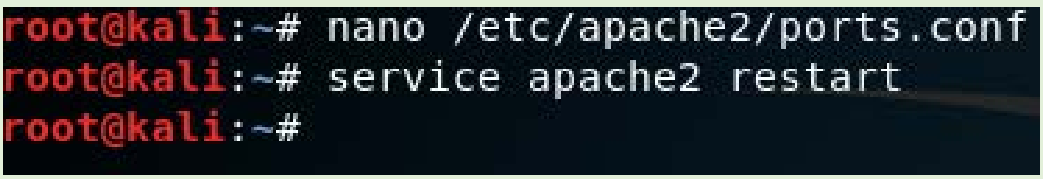


Press Enter to confirm.



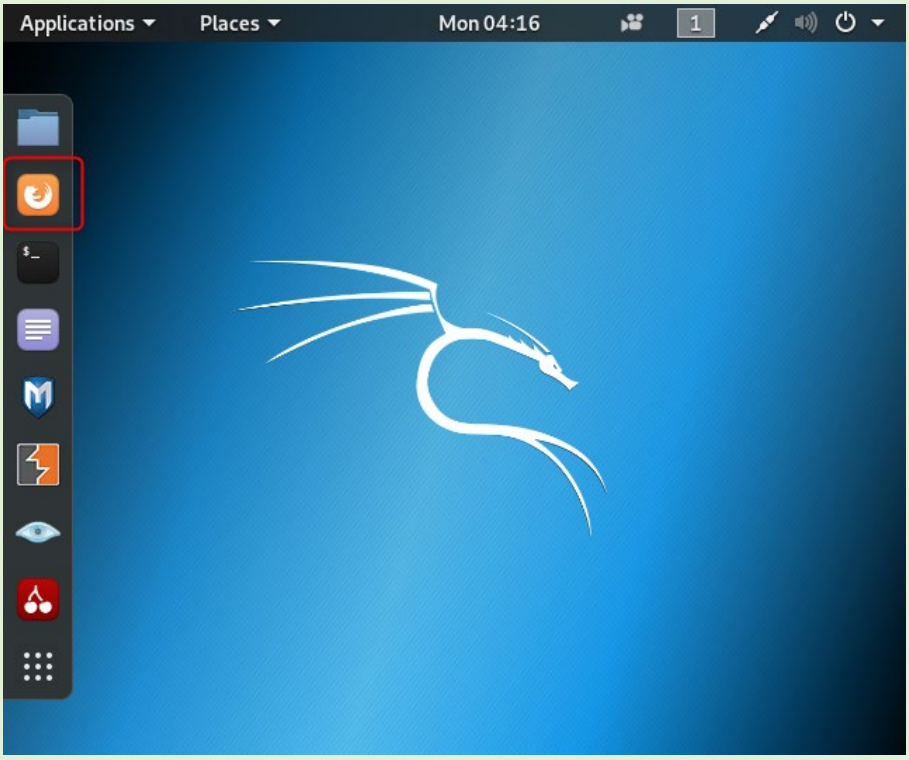
Type **service apache2 restart**, and press Enter.

Restart the service.

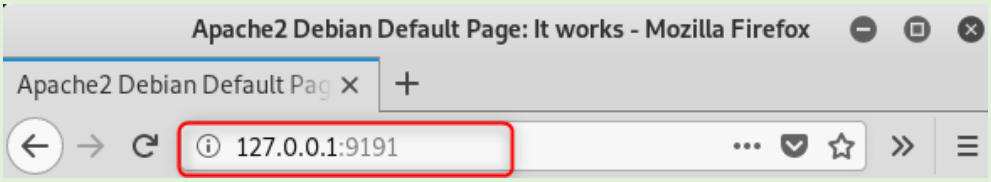


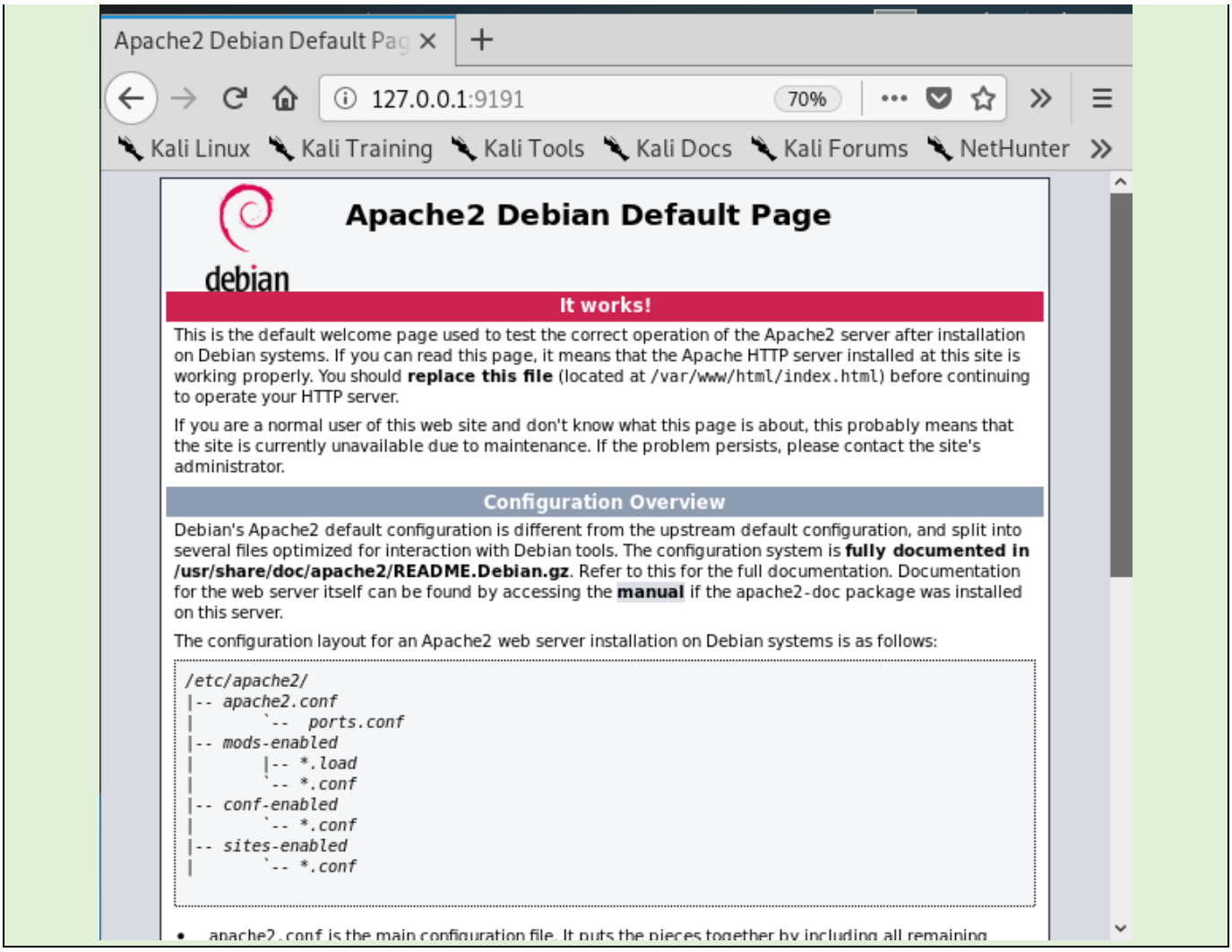
4 Browse to the website and verify that it is active.

Open the browser by clicking the Firefox icon in the menu bar.



In the browser's URL address field, type **127.0.0.1:9191**, and press Enter.



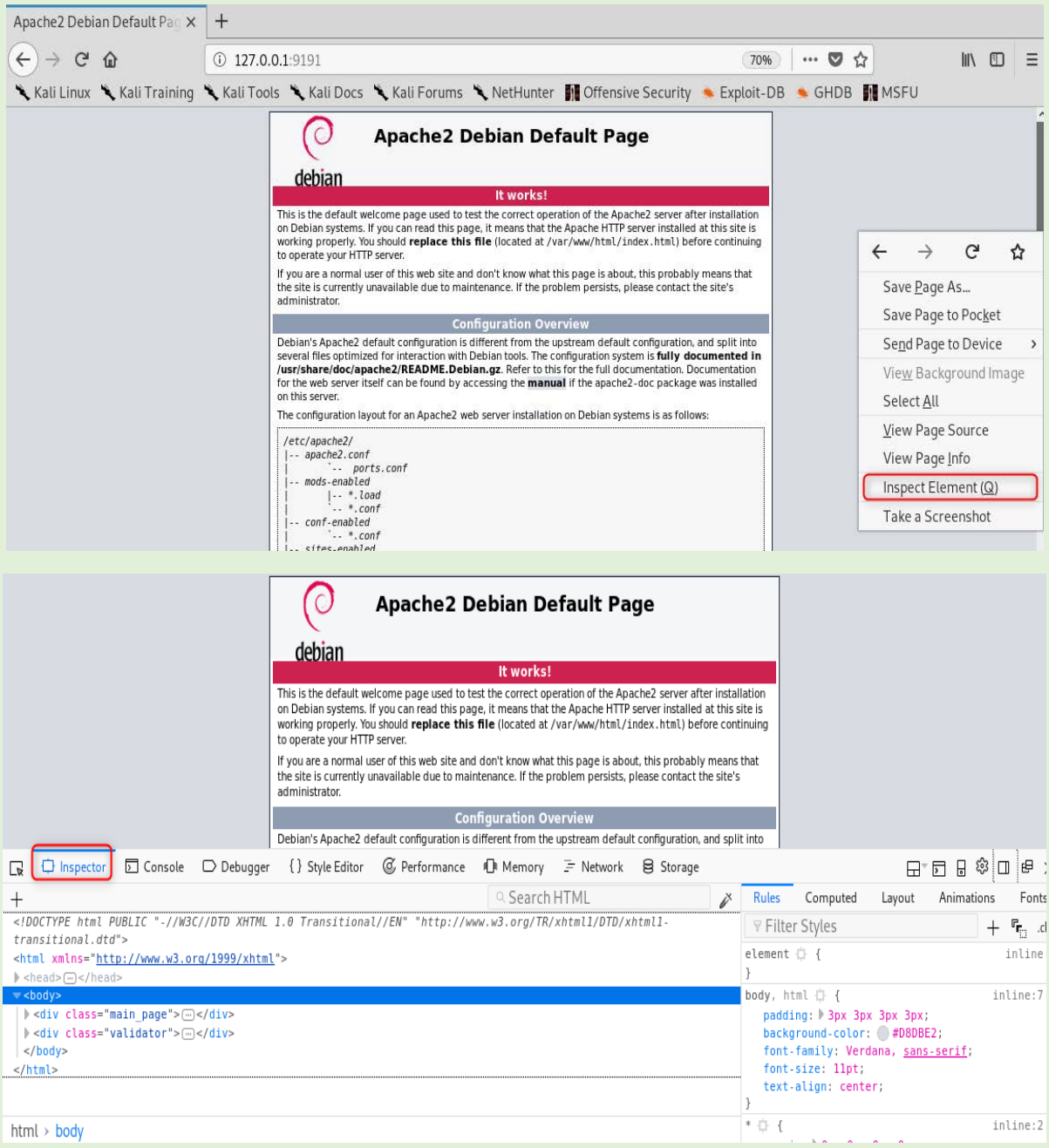

Apache2 Debian Default Pag
127.0.0.1:9191
70%
Kali Linux
Kali Training
Kali Tools
Kali Docs
Kali Forums
NetHunter
Apache2 Debian Default Page
debian
It works!
This is the default welcome page used to test the correct operation of the Apache2 server after installation on Debian systems. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should replace this file (located at /var/www/html/index.html) before continuing to operate your HTTP server.
If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.
Configuration Overview
Debian's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Debian tools. The configuration system is fully documented in /usr/share/doc/apache2/README.Debian.gz. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the manual if the apache2-doc package was installed on this server.
The configuration layout for an Apache2 web server installation on Debian systems is as follows:
/etc/apache2/
|-- apache2.conf
| `-- ports.conf
|-- mods-enabled
| |-- *.load
| `-- *.conf
|-- conf-enabled
| `-- *.conf
|-- sites-enabled
| `-- *.conf
apache2.conf is the main configuration file. It puts the pieces together by including all remaining

Lab Task 2: Enable HTTP Headers

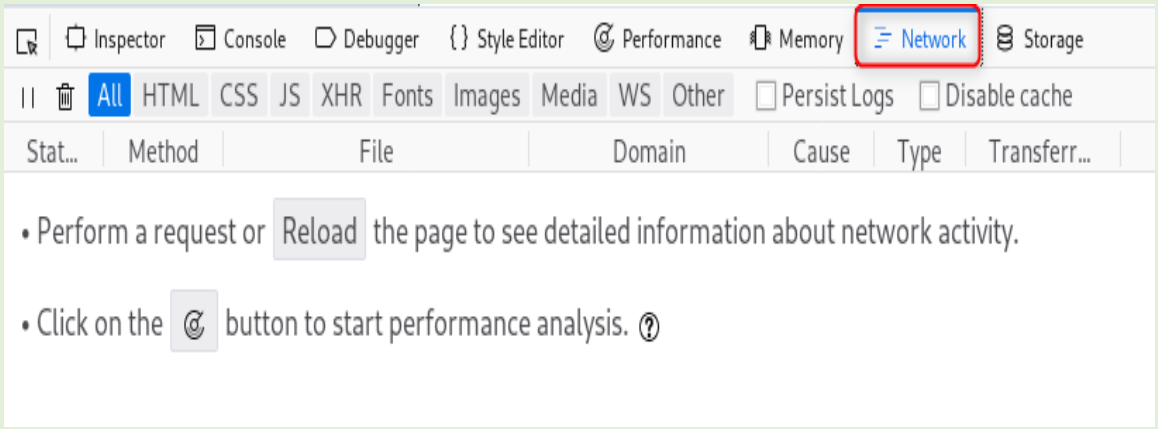
In this task, you will enable the usage of HTTP headers and create a custom header for the server.

1 Via the browser inspector, locate the content type header and examine it.

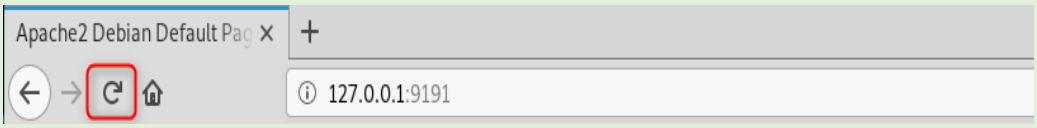
In the web page, right-click in any location of the screen, and select Inspect Element.



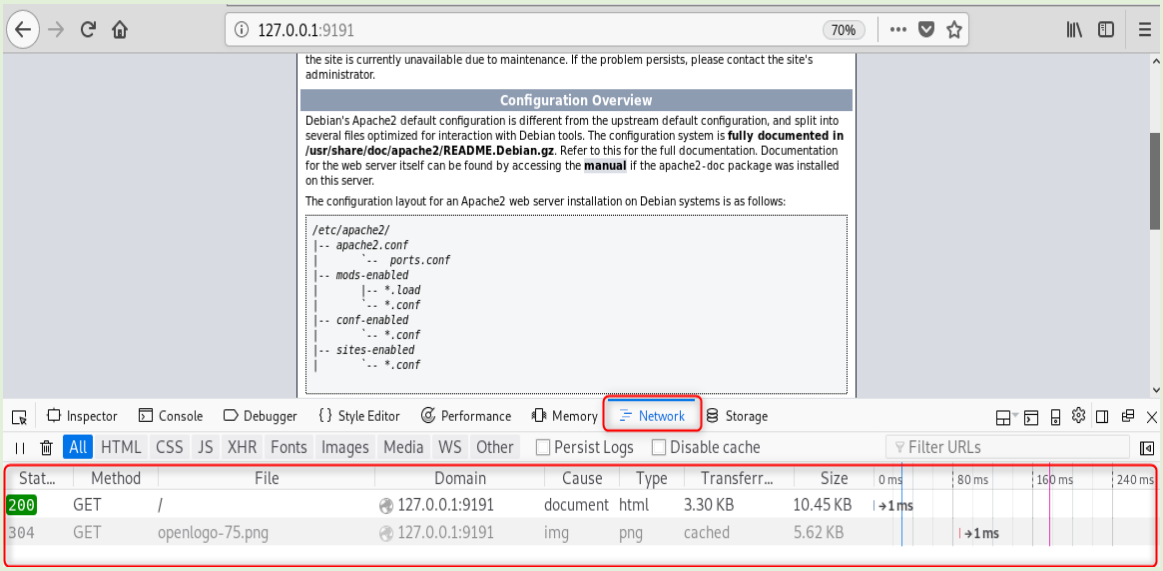
In the Inspector window, navigate to the “Network” tab:



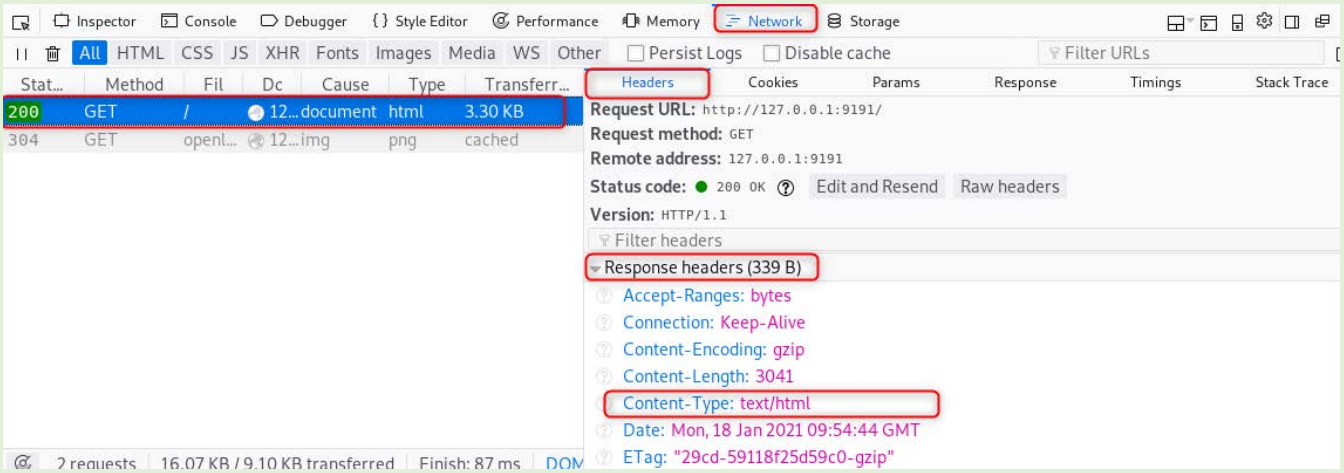
Refresh the page by clicking the icon highlighted in the example below (beside the URL address field).



After refreshing the page, click the first GET request.

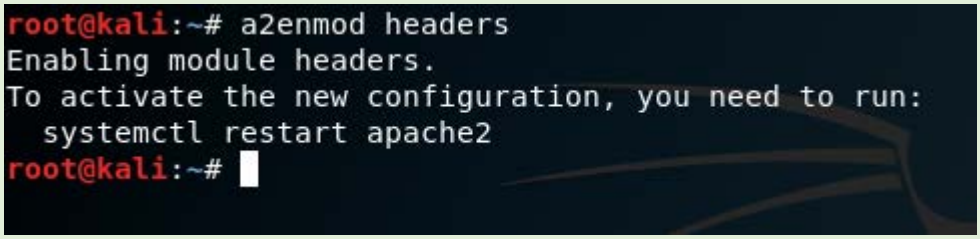


Check the response headers to detect the “Content-Type” attribute that describes the content of the page.

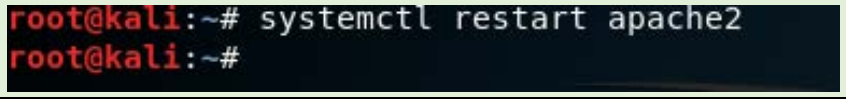


2 In the terminal, enable HTTP headers for the HTTP service.

Type **a2enmod headers** in the Kali terminal, and press Enter.



Type **systemctl restart apache2** to restart the service and activate the configuration.



Lab Task 3: Create a Custom Header

In this task, you will set a custom HTTP header for the server through the configuration files, and monitor the response header differences.

1. Open the Apache security.conf file, and add a header at the bottom using: **Header set Custom-Header: "Hello"**. Then save and inspect the changes.

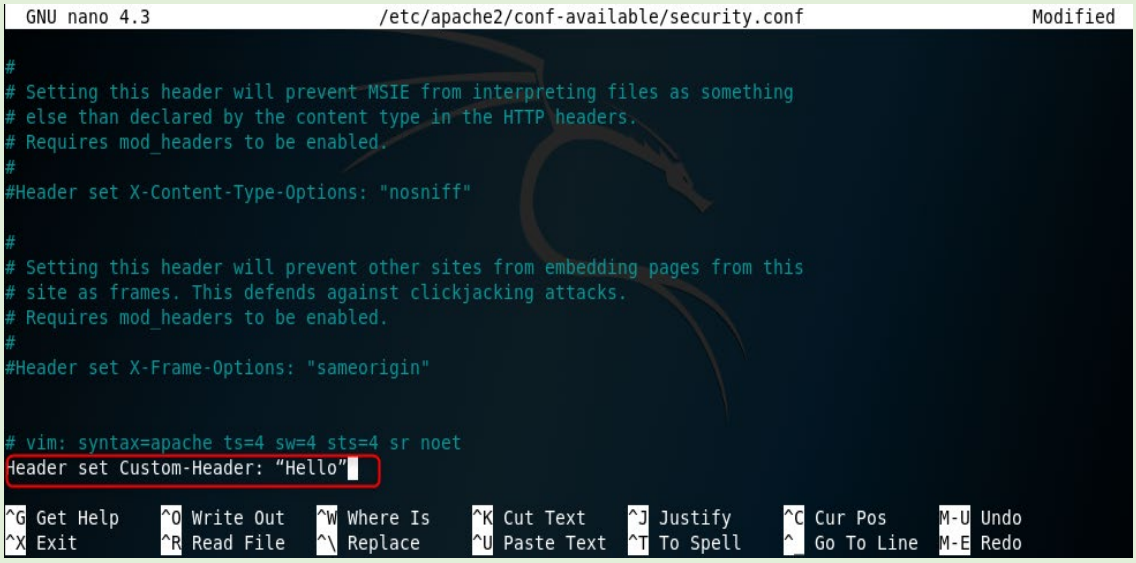
2. Make a request to the localhost to examine the change in the header.

From the terminal, run **nano /etc/apache2/conf-available/security.conf** to open the configuration file in the Nano editor.

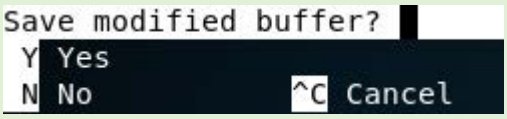
```
root@kali:~# nano /etc/apache2/conf-available/security.conf
```

Use the down arrow to scroll to the bottom of the page, and type:

Header set Custom-Header: "Hello"



Press CTRL+X to exit the Nano editor.
Type 'y' (or 'Y') to save the changes.

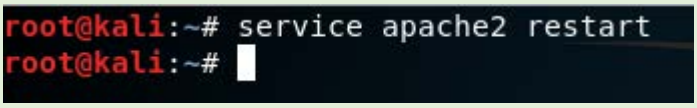


Press Enter to confirm.



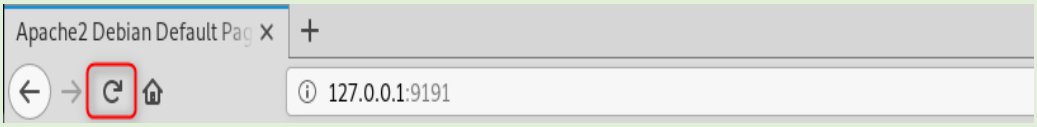
Restart the service. If the service fails to restart, something was written incorrectly in the file.

Type **service apache2 restart**, and press Enter.

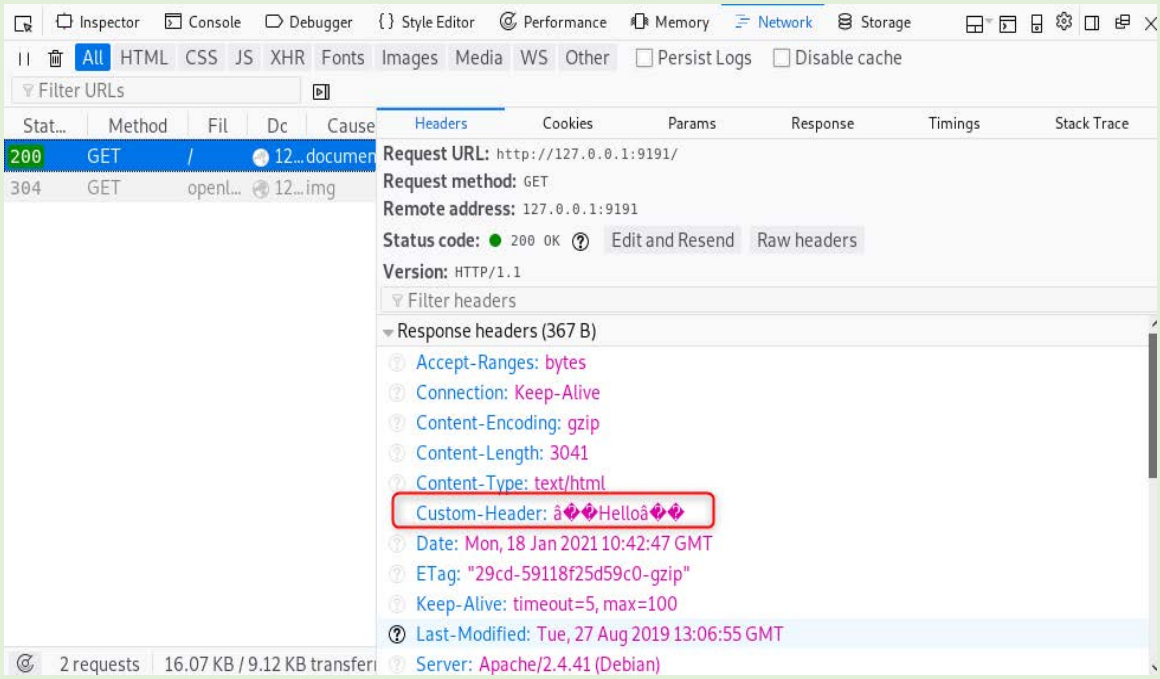


In the browser, refresh the page and inspect the newly added header.

To refresh, click the icon highlighted in the following example.



Inspect the new attributes of the first GET request.



Note the new field “Custom-Header”.